



بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ
جامعة كركري - كلية علوم الحاسوب وتقانة المعلومات
المستوى الثالث (هندسة برمجيات)
امتحان الدور الأول - ابريل 2025 م

التاريخ: 2025/4/28 م
الزمن: ثلاث ساعات

المادة: تطوير وتوثيق البرمجيات
أستاذ المادة: د. كمال بشير

Q1. True/False

- 1 Software documentation is only necessary during the development phase. () ✓
- 2 Process documentation includes user manuals and installation guides. () ✗
- 3 A use case describes interactions between users and the system. () ✓
- 4 Non-functional requirements focus on system constraints like performance. () ✗
- 5 UML is primarily used for documenting hardware architectures. ()
- 6 The SRS is part of product documentation. () ✓
- 7 The "mythical man-month" refers to delays caused by adding more developers. () ✗
- 8 User documentation includes both end-user and system administrator guides. () ✓
- 9 Scenarios are formal, structured narratives. ()
- 10 Validation ensures the system meets specified requirements. ()
- 11 The IEEE standard for user documentation requires strict adherence to all sections. () ✗
- 12 Agile projects avoid all forms of documentation. ()
- 13 A context diagram shows interactions between system components. ()
- 14 Rationale explains why design decisions were made. ()
- 15 Product documentation evolves with the software. ()
- 16 Static scheduling relies on runtime decisions. ()
- 17 Stakeholders include only developers and end-users. ()
- 18 The "design productivity gap" refers to delays in hardware development. ()
- 19 Use case diagrams are part of UML. () ✗
- 20 Architecture documentation is irrelevant for maintenance. () ✓

Q2. Define the following terms:

1. Software Documentation
2. Software Requirement Specification (SRS)
3. Process Documentation
4. Product Documentation
5. Use Case
6. Scenario
7. Hierarchical Task Analysis (HTA)
8. System Documentation

9. User Documentation
10. Functional Requirements
11. Non-Functional Requirements
12. Agile Methodology
13. UML (Unified Modeling Language)
14. Stakeholder
15. Validation
16. Verification
17. Rationale (in documentation)
18. Design Rationale
19. Change Control Board (CCB)
20. Architecture Documentation

Q3. Direct Answers

1. What is the purpose of a Software Requirements Specification (SRS)?
2. Name two types of user documentation.
3. What does UML stand for?
4. Give two examples of non-functional requirements.
5. What is the difference between validation and verification?
6. How does a scenario differ from a use case?
7. What are the three main uses of architecture documentation?
8. What is the role of a Change Control Board (CCB)?
9. Name two tools used for requirements management.
10. Why is hierarchical task analysis (HTA) useful?

Q4: Discussion/Comparison/Drawing

1. Compare process documentation and product documentation.
2. Discuss the importance of rationale in software documentation.
3. Draw a use case diagram for an online shopping system.
4. Explain how the "mythical man-month" concept impacts project management.
5. Compare functional and non-functional requirements.
6. Describe the steps involved in hierarchical task analysis (HTA).
7. Analyze the role of stakeholders in requirement gathering.
8. Discuss the challenges of maintaining up-to-date documentation in agile projects.
9. Explain the purpose of a context diagram with an example.
10. Compare static and dynamic scheduling in system design.